

ALVARO VIDELA - @old_sound

METAPHORS WE COMPUTE BY



**HOW I WENT FROM SELLING FOOD IN
THE STREET TO WORKING IN TECH**

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Computing Machinery

Hootsuite:
In Pursuit
of Reactive
Systems

IoT: The
Internet of
Terror

**Is There
a Single Method
for the Internet
of Things?**

METAPHORS WE COMPUTE BY

Complete table of contents on the following two pages

<http://queue.acm.org/detail.cfm?id=3127495>

THE YEAR IS 1980

GEORGE LAKOFF & MARK JOHNSON

METAPHORS WE LIVE BY

**METAPHOR ISN'T JUST A
MATTER OF POETRY AND
RHETORICAL FLOURISH**

METAPHORS PERMEATE ALL AREAS OF OUR LIVES

METAPHORS DICTATE

METAPHORS PERMEATE ALL AREAS OF OUR LIVES

METAPHORS DICTATE

- ▶ How we think

METAPHORS PERMEATE ALL AREAS OF OUR LIVES

METAPHORS DICTATE

- ▶ How we think
- ▶ How we behave

METAPHORS DICTATE

- ▶ How we think
- ▶ How we behave
- ▶ How we perceive

METAPHORS DICTATE

- ▶ How we think
- ▶ How we behave
- ▶ How we perceive
- ▶ How our conceptual system is built

ARGUMENT IS WAR

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- ▶ Your claims are *indefensible*

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- ▶ He *attacked every weak point* in my argument

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- ▶ I *demolished* his argument

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- ▶ I never *won* an argument with him

ARGUMENT IS WAR

- ▶ Your claims are *indefensible*
- ▶ He *attacked every weak point* in my argument
- ▶ I *demolished* his argument
- ▶ I never *won* an argument with him
- ▶ His criticisms were *right on target*

**WHAT IF ARGUMENT
IS A DANCE?**

I'M NOT CONVINCED

**LET'S TALK ABOUT
POLITICS**

HOW METAPHORS SHAPE WOMEN'S LIVES

Consider an experiment that explored how the metaphors of crime can affect people's decision-making. In 2011, Lera Boroditsky and Paul H Thibodeau at Stanford University asked students to read one of two crime reports; one described crime as a "wild beast preying on the city" and the other as a "virus infecting the city". The solutions that the students presented to reduce crime were fascinating: 75% of the 'beast' students thought jail or punishment would resolve crime and 25% suggested social reforms. Yet of those that had been told crime "plagued" neighbourhoods, only 56% opted for more enforcement and 44% wanted social reforms.

BEAST VS PLAGUE

FEMINISM CONFRONTS TECHNOLOGY

J U D Y
W A I C M A N



are referred to by the female pronoun! Similarly, the complementary values of hard/soft are also used to legitimate female exclusion from the world of engineering.⁸ Masculinity is expressed both in terms of muscular physical strength and aggression, and in terms of analytical power. 'At one moment, in order to fortify their identification with physical engineering, men dismiss the intellectual world as "soft". At the next moment, however, they need to appropriate sedentary, intellectual engineering work for masculinity too.' (Cockburn, 1985, p. 190)

No matter how masculinity is defined according to this ever-adaptable ideology, it always constructs women as ill-suited to technological pursuits.

**I'M STILL NOT
CONVINCED**

HUMAN RESOURCE MANAGEMENT

PEOPLE ARE NOT RESOURCES



**TRIGGER
WARNING**

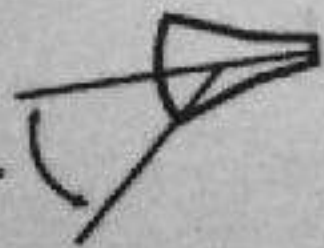
**GIVING A PLATFORM
TO RACISTS**

**“WRESTLING WITH
INCLUSION AT XYZCONF”**

**“WRESTLING WITH
INCLUSION AT XYZCONF”**

**LET'S TALK ABOUT
COMPUTERS**

40°, ± 7°

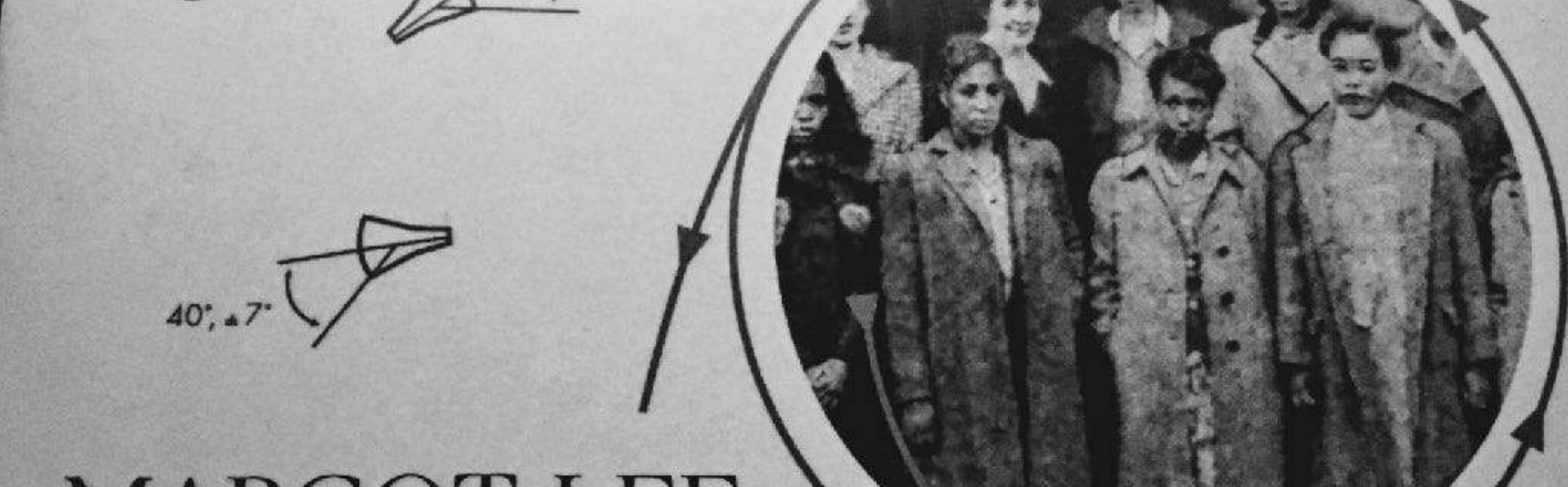


MARGOT LEE
SHETTERLY



HIDDEN +

ENCLOSURES



COMPUTERS

HIDDEN +
INCLINED

**METAPHORS ENABLE
UNDERSTANDING**



JULIET IS LIKE THE SUN

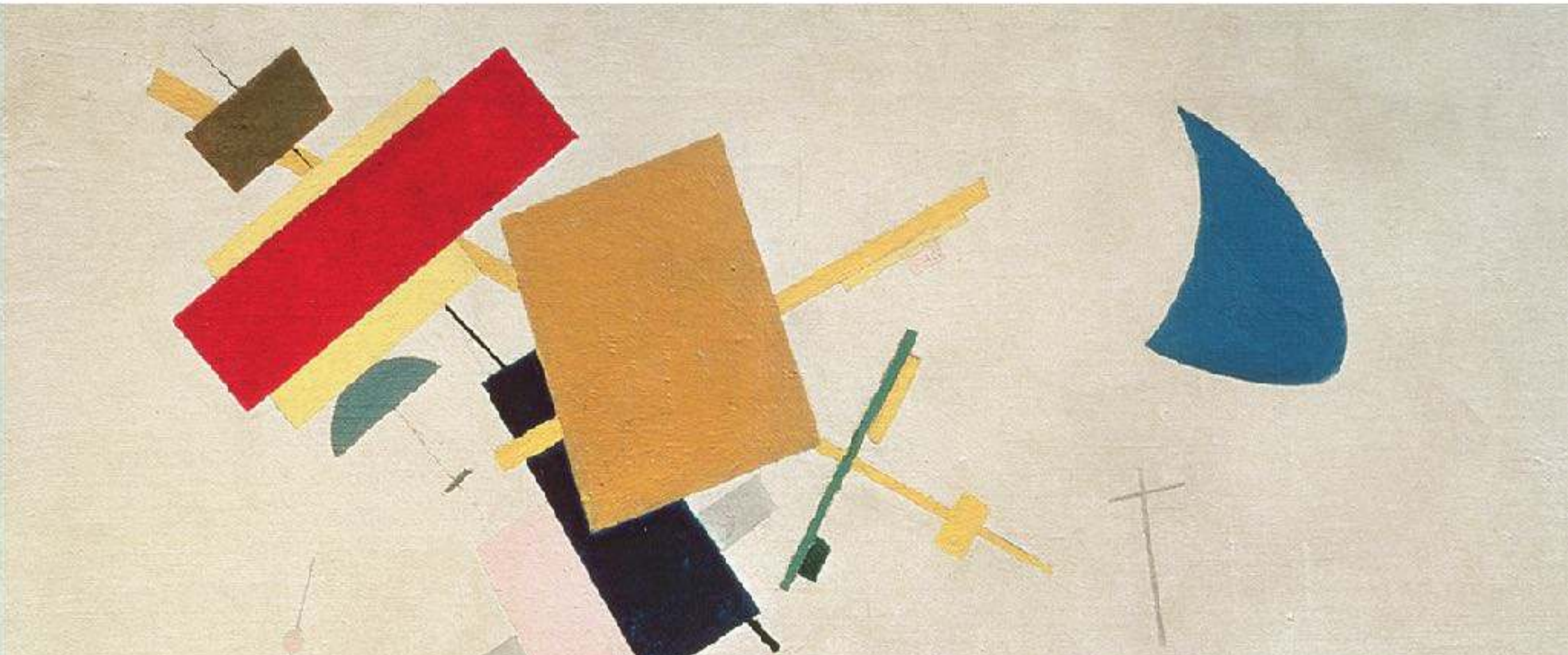


**JULIET GAVE ME
SKIN CANCER**

THE GEOMETRY OF MEANING

SEMANTICS BASED ON CONCEPTUAL SPACES

PETER GÄRDENFORS



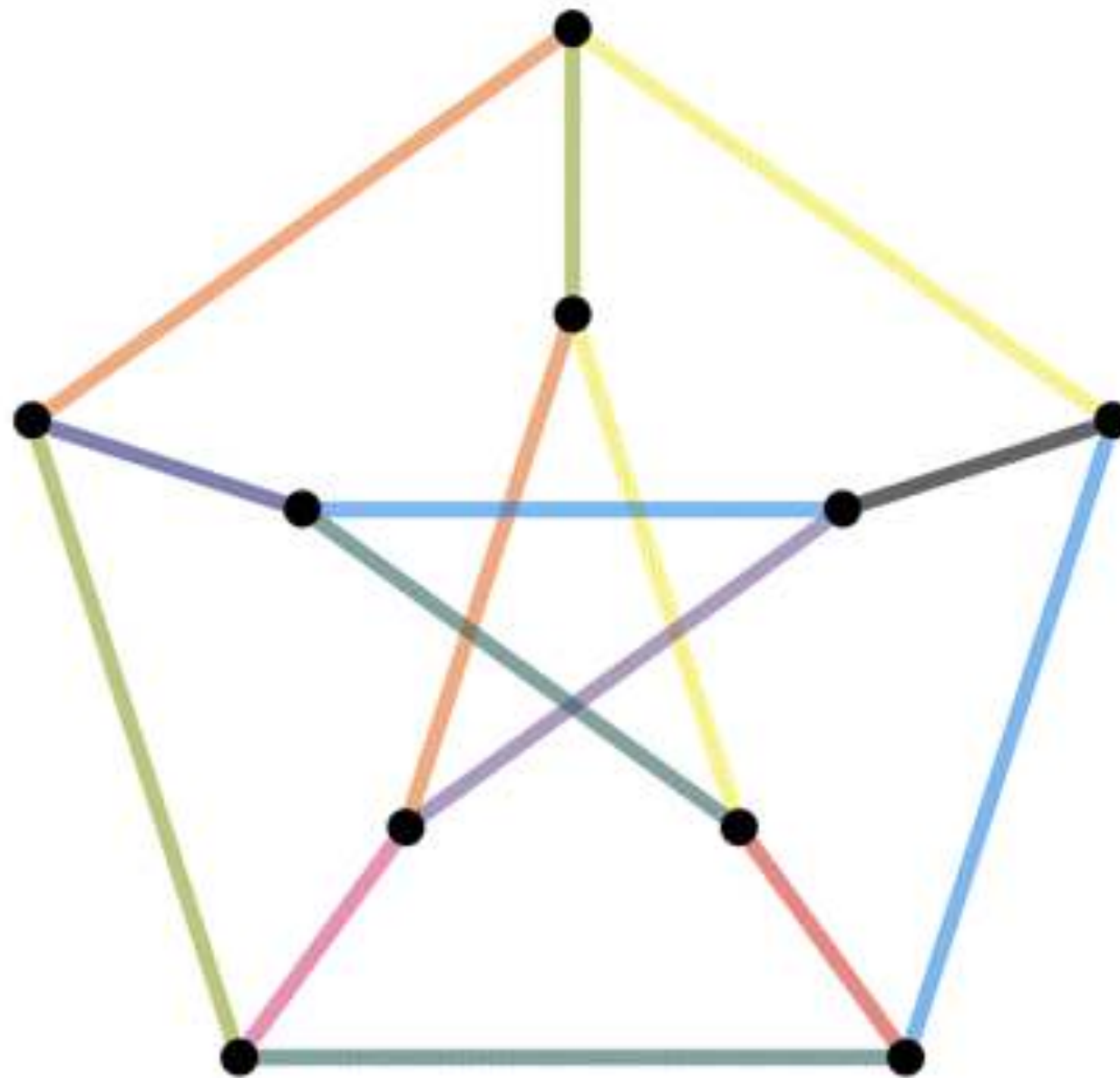
**METAPHORICAL
MAPPINGS PRESERVE THE
THE COGNITIVE TOPOLOGY
OF THE SOURCE DOMAIN**

**IN A WAY CONSISTENT
WITH THE INHERENT
STRUCTURE OF THE
TARGET DOMAIN**

**METAPHORS TRANSFER
INFORMATION FROM
ONE CONCEPTUAL
DOMAIN TO ANOTHER**

**WHAT IS TRANSFERRED
IS A PATTERN RATHER
THAN DOMAIN
SPECIFIC INFORMATION**

**A METAPHOR CAN THUS BE
USED TO IDENTIFY A
STRUCTURE IN A DOMAIN
THAT WOULD NOT HAVE BEEN
DISCOVERED OTHERWISE**



GRAPH ISOMORPHISM

<https://www.quantamagazine.org/algorithm-solves-graph-isomorphism-in-record-time-20151214>

**THIS IS HOW
METAPHORS CREATE
NEW KNOWLEDGE**

**METAPHORS OBSCURE
UNDERSTANDING**



TELE-GRAPH

**“SOMETIMES OUR TOOLS DO WHAT WE
TELL THEM TO. OTHER TIMES, WE
ADAPT OURSELVES TO OUR TOOLS’
REQUIREMENTS”**

Nicholas Carr

**METAPHORS ARE THE
TOOLS OF THOUGHT**

METAPHORS AND CODE

WHAT A PROGRAMMER DOES

It has been believed that a programmer occasionally writes code and gets it running on a computer, and that this is what he is paid for. In spite of his obvious inefficiency, no one else seems to do this work more effectively. However, his activity is still observed principally as loafing—a kind of ritual (like the British and teatime) which must be put up with.

Another view of what a programmer does addresses more constructively all that “wasted” time and

cludes more than the running code, more than the symbolic code, or even the operator’s guide, the maintenance guide, or the design guide. For in fact, in response to any serious breach of the program’s integrity, a programmer will become involved, as part of the integral organization built by the original programmer. If one now looks closely, he can begin to recognize the intent of those steps in the ritual of programming.

WHAT A PROGRAMMER DOES

It has been believed that a programmer occasionally writes code and gets

cludes more than the running code, more than the symbolic code, or even

BEST UNKNOWN PAPER

company as running a kind of model (like the British and teatime) which must be put up with.

Another view of what a programmer does addresses more constructively all that "wasted" time and

part of the integral organization built by the original programmer. If one now looks closely, he can begin to recognize the intent of those steps in the ritual of programming.

**“TO PROGRAM IS TO WRITE TO
ANOTHER PROGRAMMER
ABOUT OUR SOLUTION TO A PROBLEM”**

What a Programmer Does

**“PROGRAMS MUST BE WRITTEN FOR
PEOPLE TO READ, AND ONLY
INCIDENTALLY FOR MACHINES TO
EXECUTE”**

Structure and Interpretation of Computer Programs

THE USE OF SUB-ROUTINES IN PROGRAMMES

D. J. Wheeler

Cambridge & Illinois Universities

THE USE OF SUB-ROUTINES IN PROGRAMMES

D. J. Wheeler

Cambridge & Illinois Universities

The above remarks may be summarized by saying sub-routines are very useful-although not absolutely necessary-and that the prime objectives to be born in mind when constructing them are simplicity of use, correctness of codes and accuracy of description. All complexities should-if possible-be buried out of sight.



The library of tapes on which subroutines are punched is contained in the steel cabinet shown on the left. The operator is punching a program tape on keyboard perforator. By placing library tapes on the tapereader shown in the center of the photograph, the operator can copy them mechanically onto the tape she is preparing.

METAPHORS AND CODE

**TYPES ARE THE CHARACTERS
THAT TELL THE STORY OF
OUR PROGRAMS**

PROGRAMMING WITH ABSTRACT DATA TYPES

Barbara Liskov
Massachusetts Institute of Technology
Project MAC
Cambridge, Massachusetts

PROGRAMMING WITH ABSTRACT DATA TYPES

The motivation behind the work in very-high-level languages is to ease the programming task by providing the programmer with a language containing primitives or abstractions suitable to his problem area. The programmer is then able to spend his effort in the right place; he concentrates on solving his problem, and the resulting program will be more reliable as a result. Clearly, this is a worthwhile goal.

Unfortunately, it is very difficult for a designer to select in advance all the abstractions which the users of his language might need. If a language is to be used at all, it is likely to be used to solve problems which its designer did not envision, and for which the abstractions embedded in the language are not sufficient.

This paper presents an approach which allows the set of built-in abstractions to be augmented when the need for a new data abstraction is discovered. This approach to the handling of abstraction is an outgrowth of work on designing a language for structured programming. Relevant aspects of this language are described, and examples of the use and definitions of abstractions are given.

**WITHOUT TYPES WE JUST
HAVE OPERATIONS ON
STREAM OF BYTES**

CHOOSING THE RIGHT DATA STRUCTURE

CHOOSE THE RIGHT DATA STRUCTURE

CHOOSE THE RIGHT DATA STRUCTURE

- ▶ Array

CHOOSE THE RIGHT DATA STRUCTURE

- ▶ Array

- ▶ Set

CHOOSE THE RIGHT DATA STRUCTURE

- ▶ Array
- ▶ Set
- ▶ LinkedList

CHOOSE THE RIGHT DATA STRUCTURE

- ▶ Array
- ▶ Set
- ▶ LinkedList
- ▶ Queue

CHOOSE THE RIGHT DATA STRUCTURE

- ▶ Array
- ▶ Set
- ▶ LinkedList
- ▶ Queue
- ▶ Stack

**A PROGRAM'S EXPLANATORY
POWER IS THE MEASURE OF
ITS OWN ELEGANCE**

**DATA STRUCTURES
HAVE EXPLANATORY
POWER**

COGNITIVE LEAPS

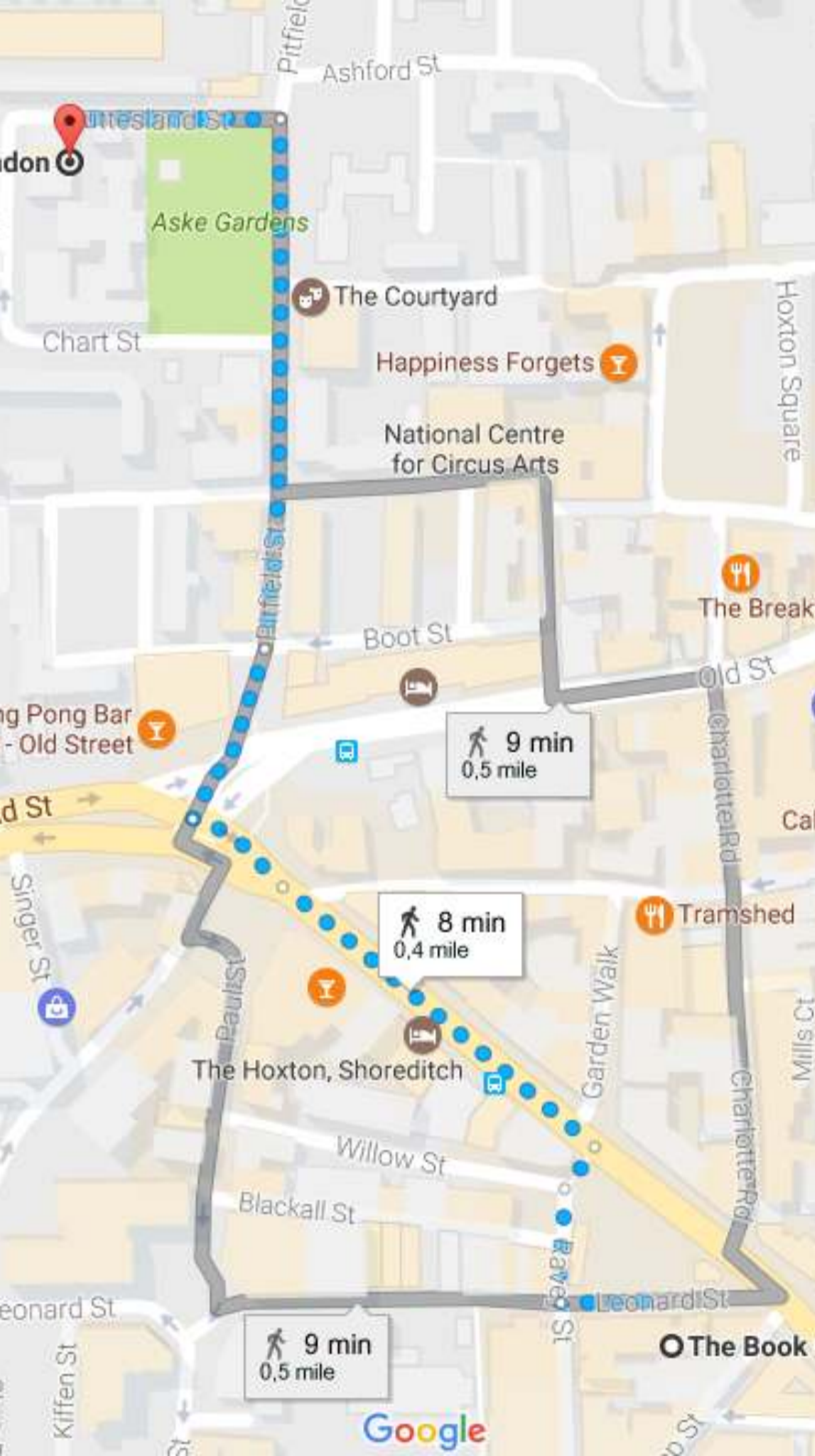


TASK SCHEDULING

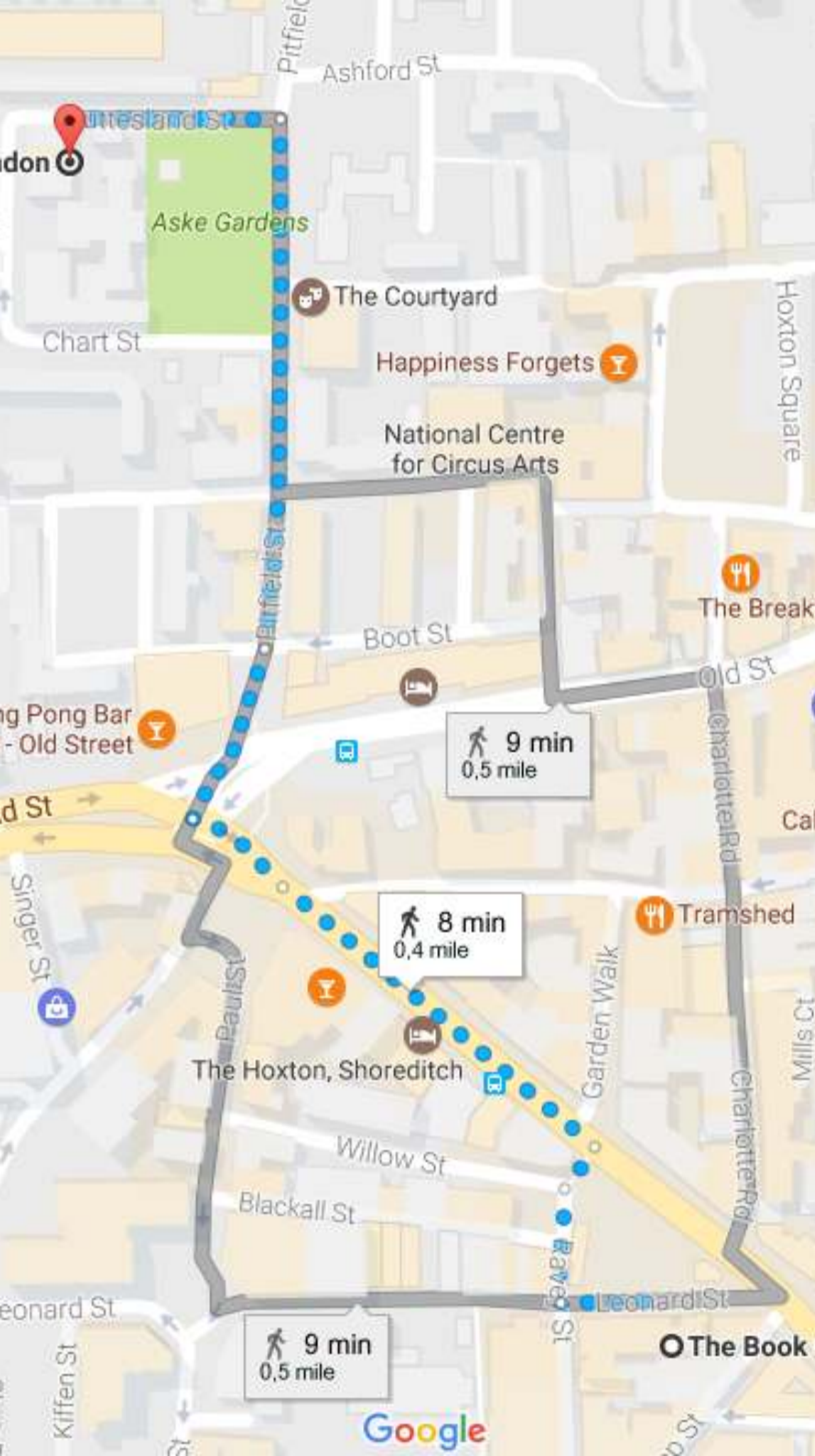


TASK SCHEDULING

QUEUEING
THEORY



ROUTE PLANNING



ROUTE PLANNING

GRAPH THEORY



DATABASE REPLICATION



DATABASE
REPLICATION

**RUMOUR
MONGERING**

THE MATHEMATICAL THEORY OF EPIDEMICS

NORMAN T. J. BAILEY, M.A.

Reader in Biometry, University of Oxford;
Formerly Statistician to the Medical School,
University of Cambridge



LONDON
CHARLES GRIFFIN & COMPANY LIMITED

DATABASE REPLICATION

EPIDEMICS

**SO EVERYTHING IS A
METAPHOR?**

I DON'T BELIEVE YOU

DISTRIBUTED SYSTEMS METAPHORS

Whenever *nodes* need to *agree* on a common value, we start a *consensus* algorithm to *decide* on a value.

There's usually a *leader* process that takes care of making the final decision based on the *votes* it has received from its *peers*.



SEVEN METAPHORS

DISTRIBUTED SYSTEMS METAPHORS

Nodes communicate sending *messages* over a *channel*, which might get *congested* due to *too much traffic*. This could create an information *bottleneck*, with *queues* at each end of the *channels* backing up.

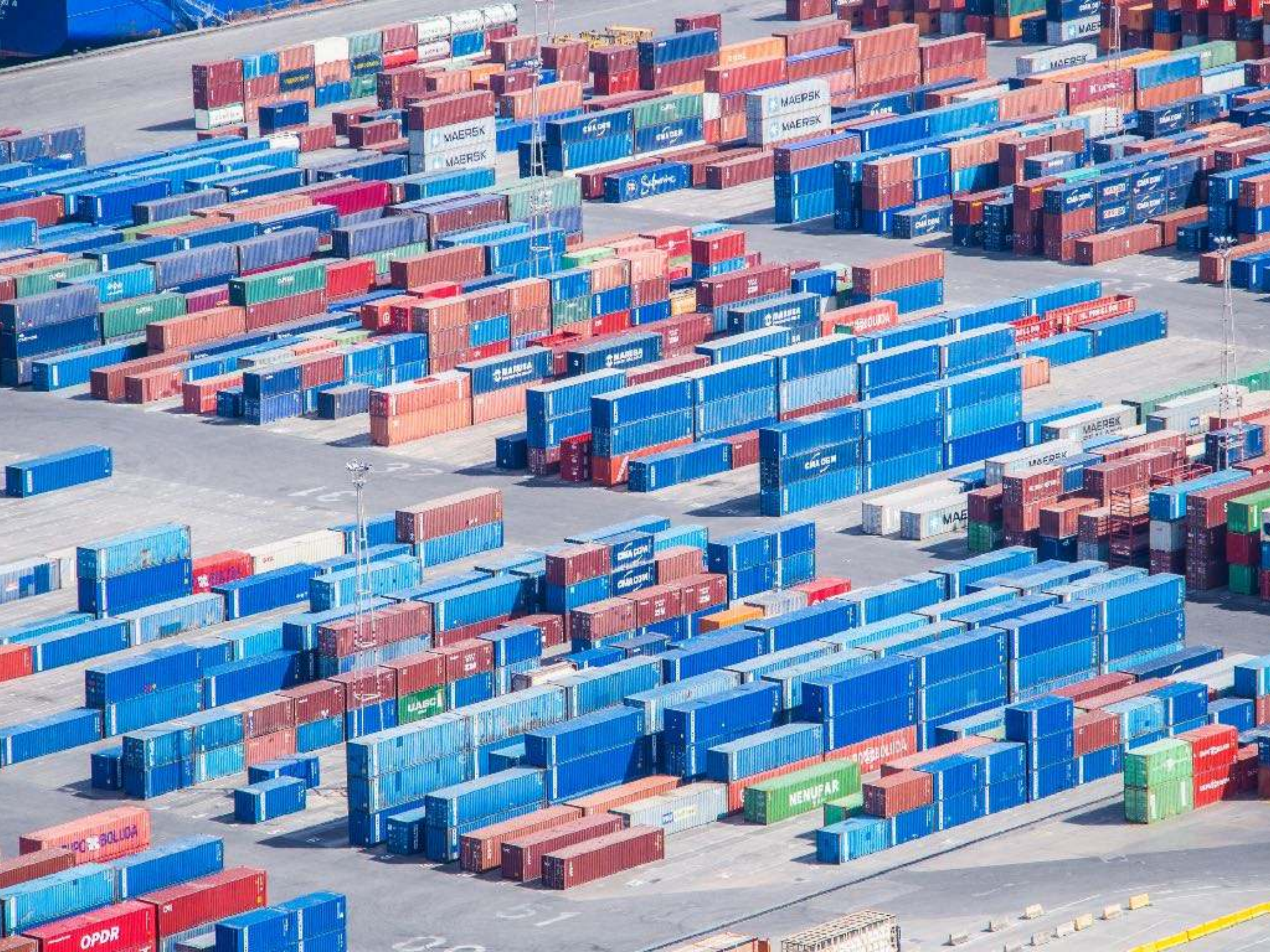


DISTRIBUTED SYSTEMS METAPHORS

These ***bottlenecks*** might render one or more nodes **unresponsive**, causing ***network partitions***. Is the process that's taking too long to ***respond dead***? We won't know unless we set a timeout...



BUZZWORDS

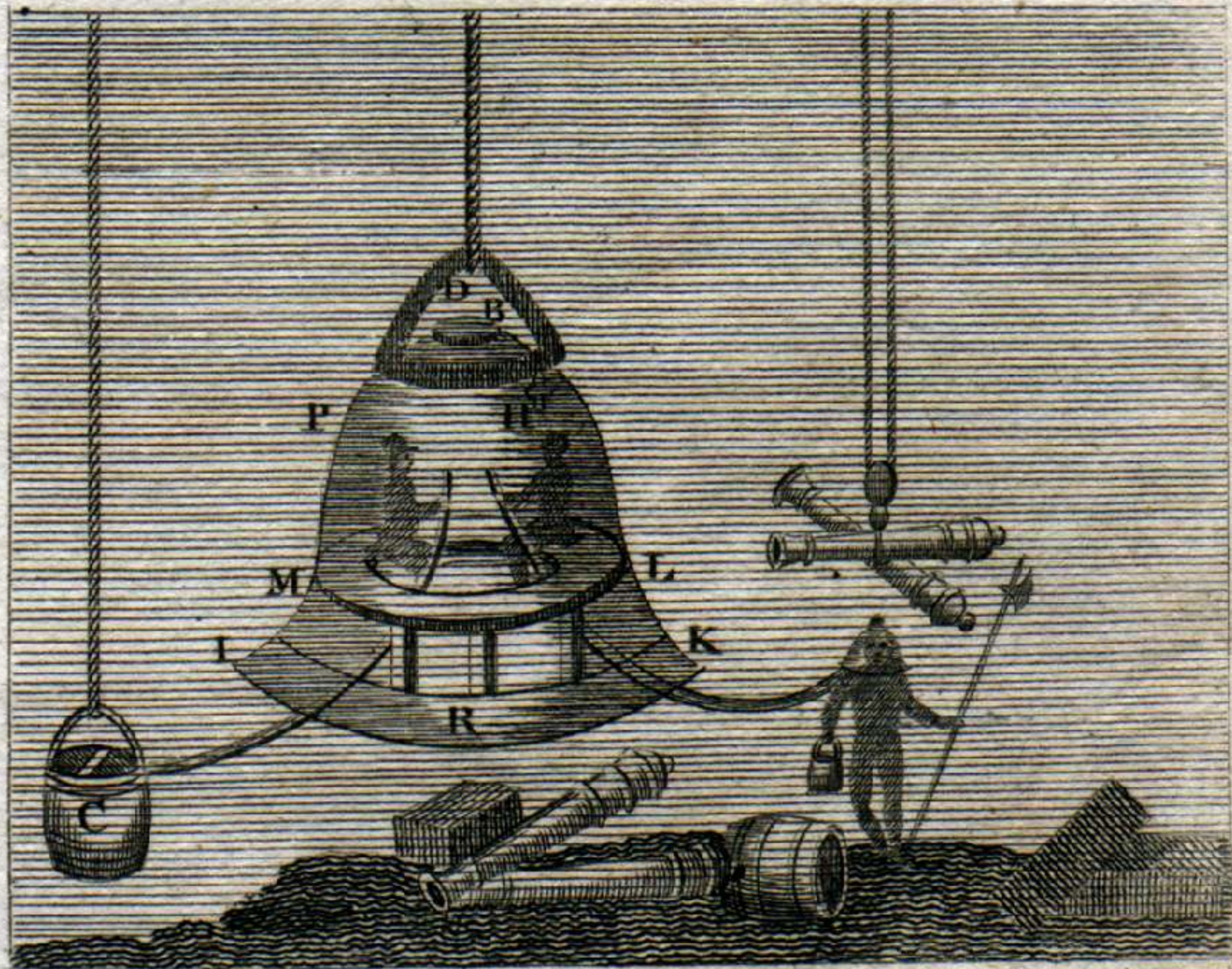


CONTAINERS

- ▶ Standard
- ▶ Ship Anywhere
- ▶ Train, Ships, Trucks
- ▶ Stackable
- ▶ Reusable



Halley's Diving Bell.



Microservices

a definition of this new architectural term

MICROSERVICES

25 March 2014



James Lewis

James Lewis is a Principal Consultant at ThoughtWorks and member of the Technology Advisory Board. James'

interest in building applications out of small collaborating services stems from a background in integrating enterprise systems at scale. He's built a number of

Contents

Characteristics of a Microservice Architecture

Componentization via Services

Organized around Business Capabilities

Products not Projects

Smart endpoints and dumb pipes

Decentralized Governance

Decentralized Data Management

Infrastructure Automation

Design for failure

Evolutionary Design

MICROSERVICES

- ▶ Decentralised Governance
- ▶ Monolith vs. Microservice
- ▶ Isolation
- ▶ Collaboration
- ▶ Small is better - Big is cumbersome
- ▶ David vs. Goliath

ERLANG ANYONE?

“IN ANOTHER DIRECTION, ONE COULD ARGUE THAT MICROSERVICES ARE THE SAME THING AS THE ERLANG PROGRAMMING MODEL, BUT APPLIED TO AN ENTERPRISE APPLICATION CONTEXT”

**WHAT'S ERLANG'S
ELEVATOR PITCH?**

**FIRST GET PEOPLE TO
UNDERSTAND THINGS**

**THEN EXPLAIN HOW
THINGS ACTUALLY WORK**

RABBITMQ

A JOB SERVER?

CONCLUSION

MASTER THE ART OF METAPHOR SELECTION

**MASTER THE ART OF
MEANING AMPLIFICATION**

**OUR PROGRAM IS THE
METAPHOR FOR THE
SOLUTION WE FOUND**

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- ▶ Office Workers: <https://flic.kr/p/5WwpeV>
- ▶ Sun: <https://flic.kr/p/9Q6SY1>
- ▶ Queue: <https://flic.kr/p/8AqWW7>
- ▶ Consensus: <https://flic.kr/p/aws7dH>
- ▶ Bottlenecks: <https://flic.kr/p/EJ5Q3>
- ▶ Gossip: <https://flic.kr/p/4bCDr2>
- ▶ Containers: <https://flic.kr/p/nWLQxE>

THANK YOU!

@old_sound